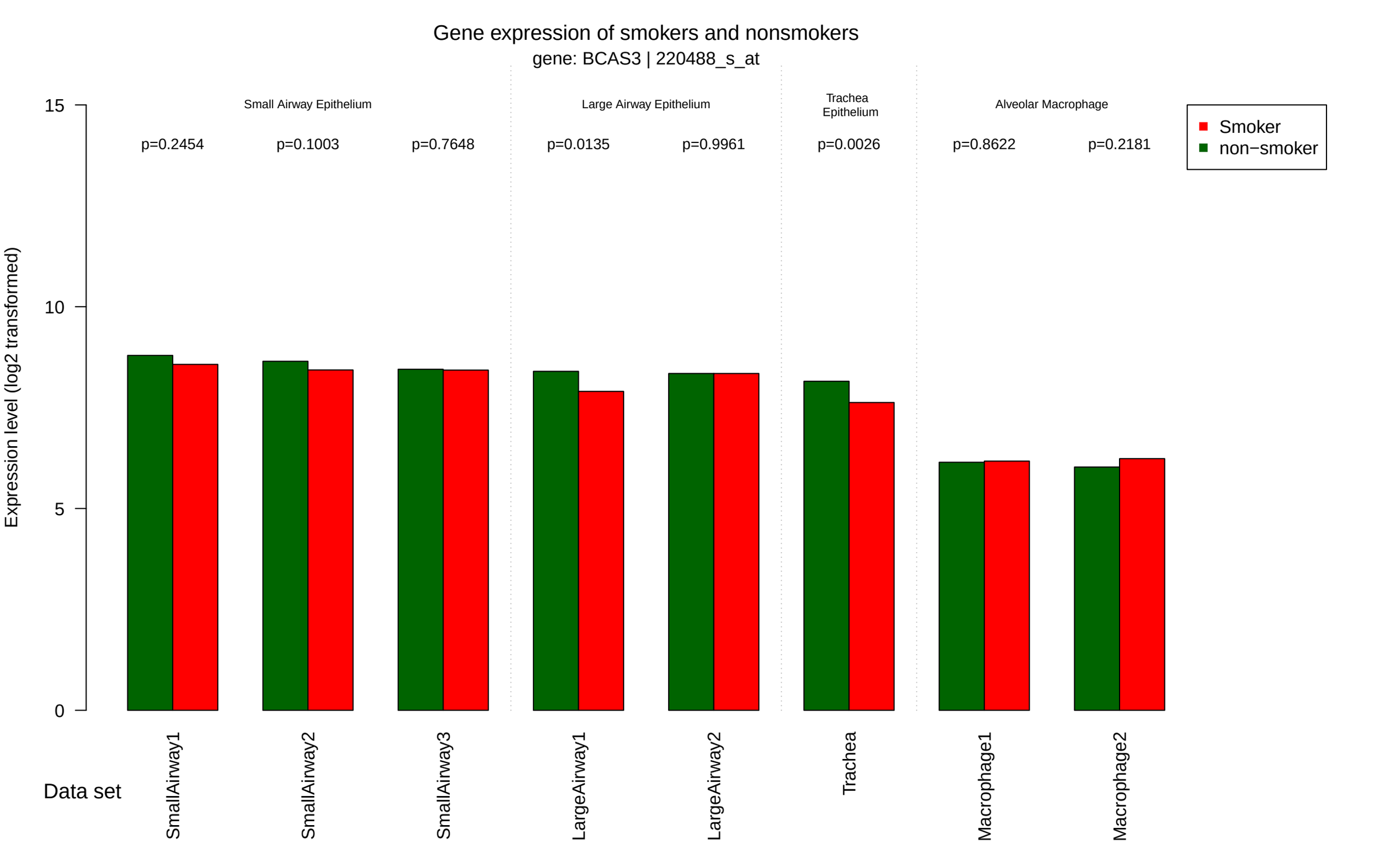
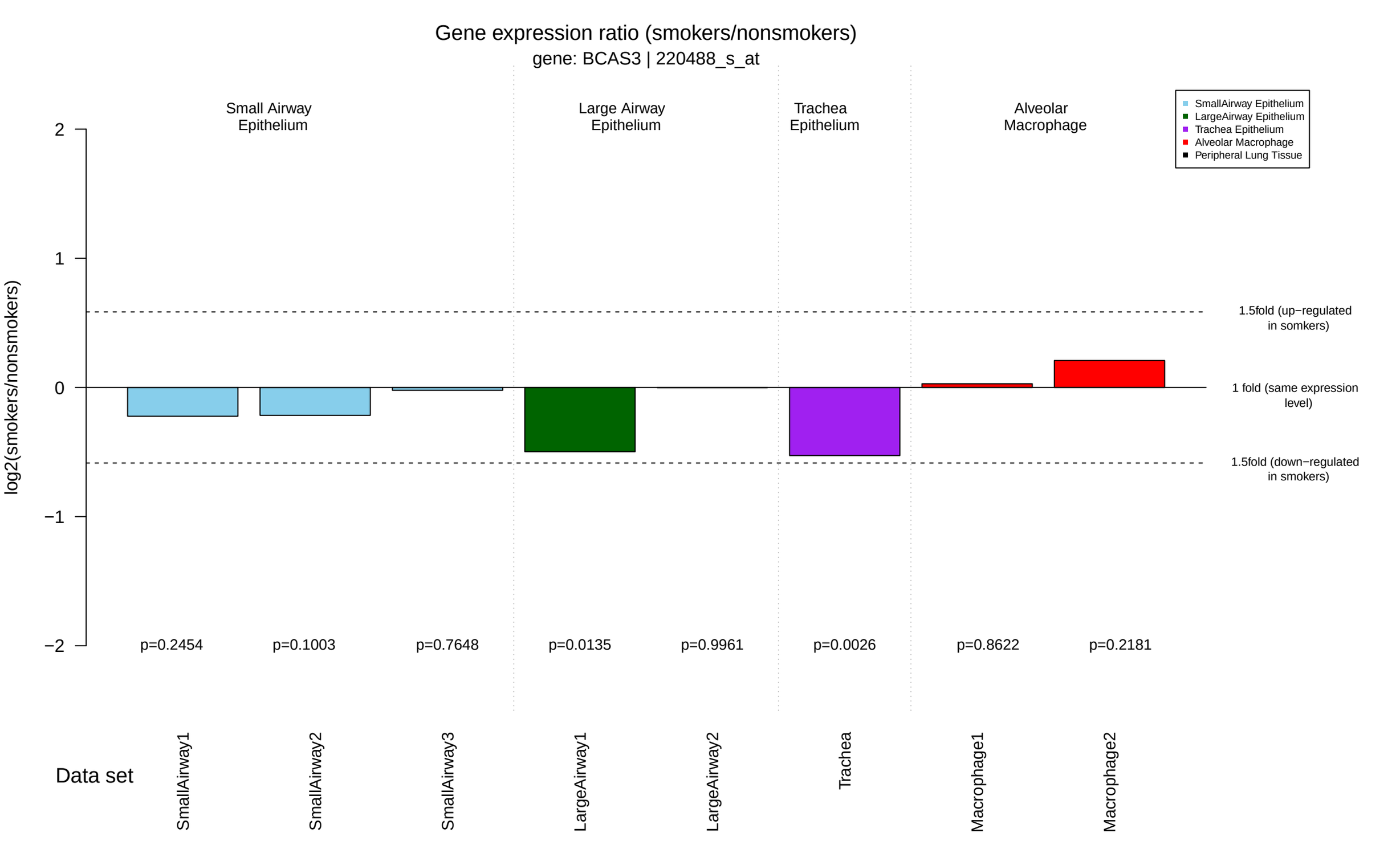
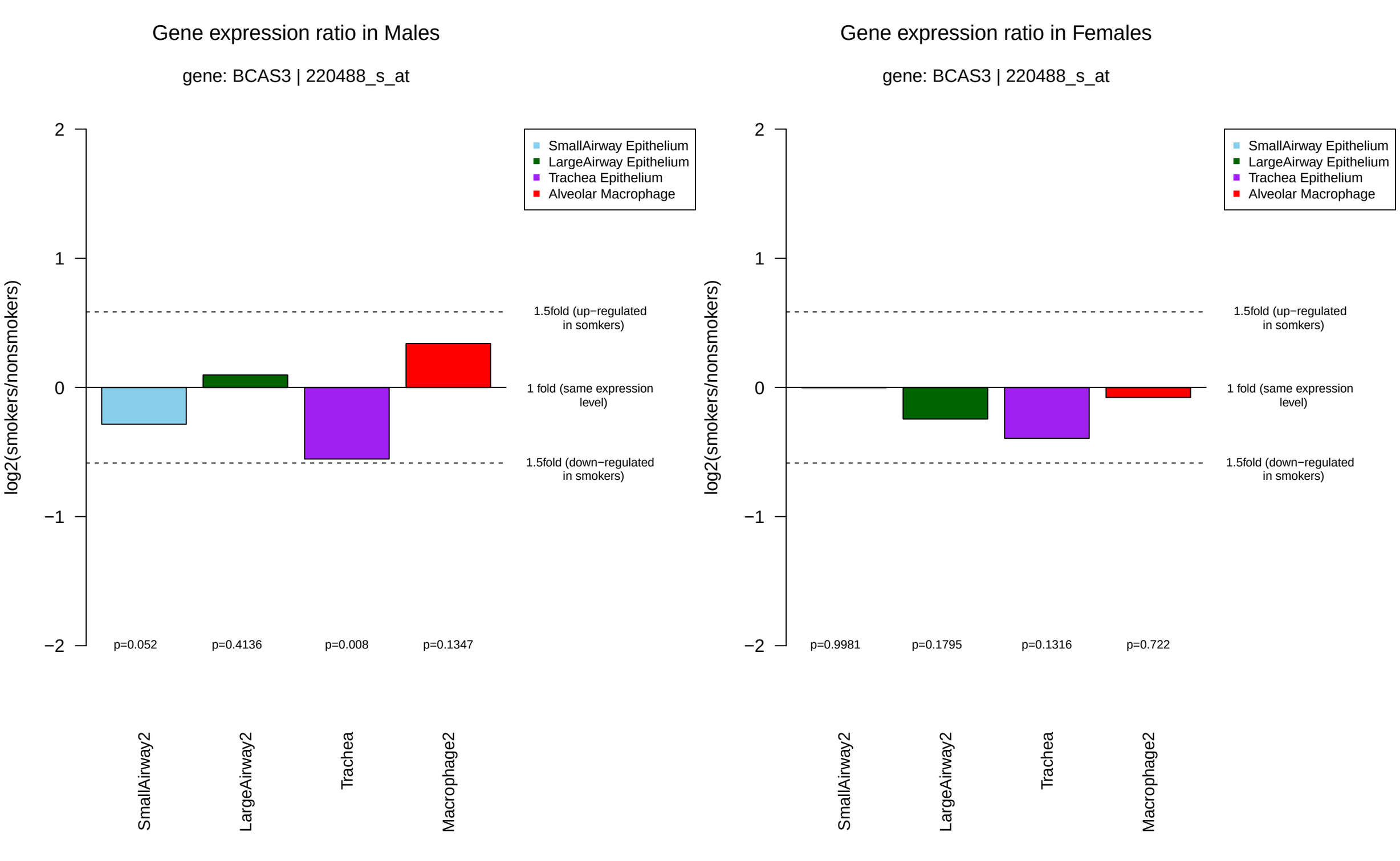
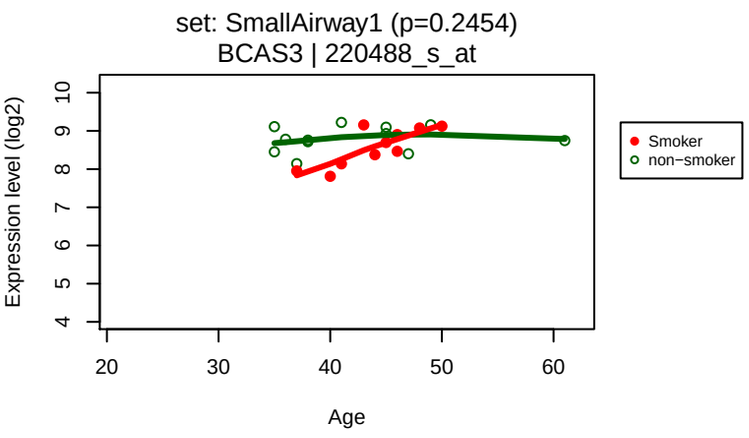
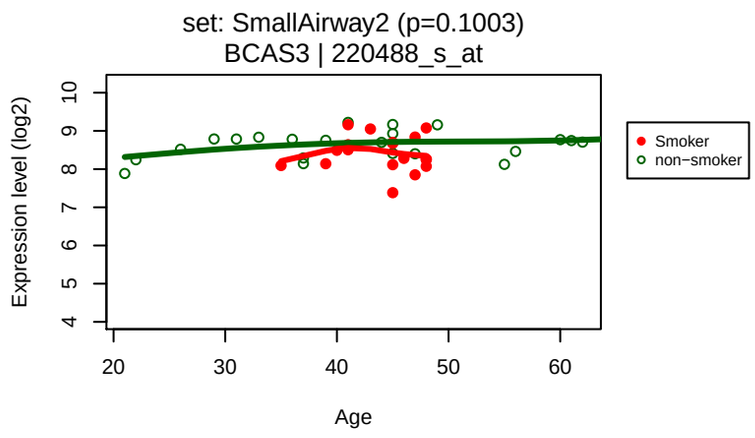
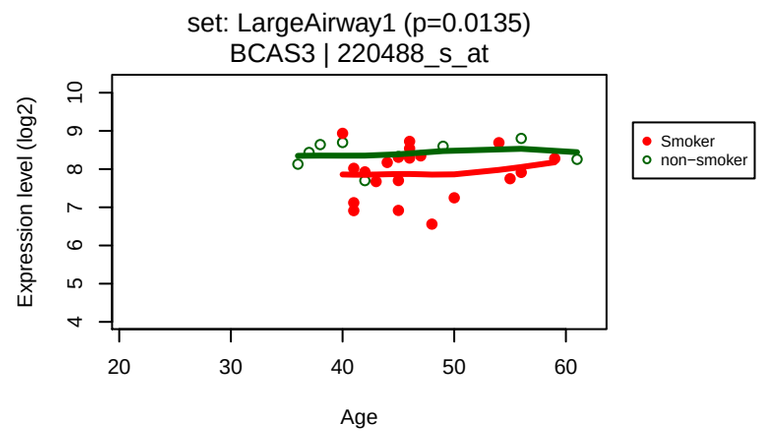

Gene expression of smokers and nonsmokers
gene: BCAS3 | 220488_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2454
p=0.1003
p=0.7648
p=0.0135
p=0.9961
p=0.0026
p=0.8622
p=0.2181
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

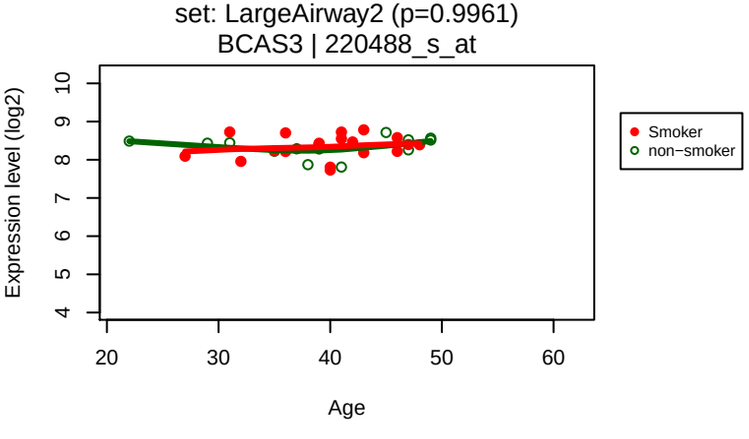

Gene expression ratio (smokers/nonsmokers)
gene: BCAS3 | 220488_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2454
p=0.1003
p=0.7648
p=0.0135
p=0.9961
p=0.0026
p=0.8622
p=0.2181
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

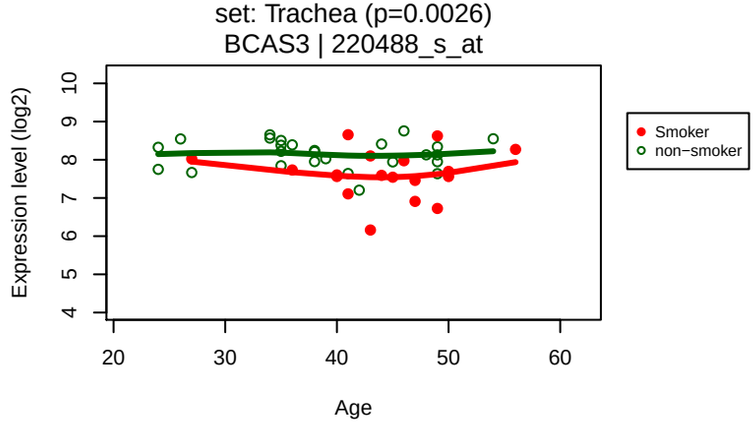

Gene expression ratio in Males
gene: BCAS3 | 220488_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.052
p=0.4136
p=0.008
p=0.1347
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: BCAS3 | 220488_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9981
p=0.1795
p=0.1316
p=0.722
SmallAirway2
LargeAirway2
Trachea
Macrophage2

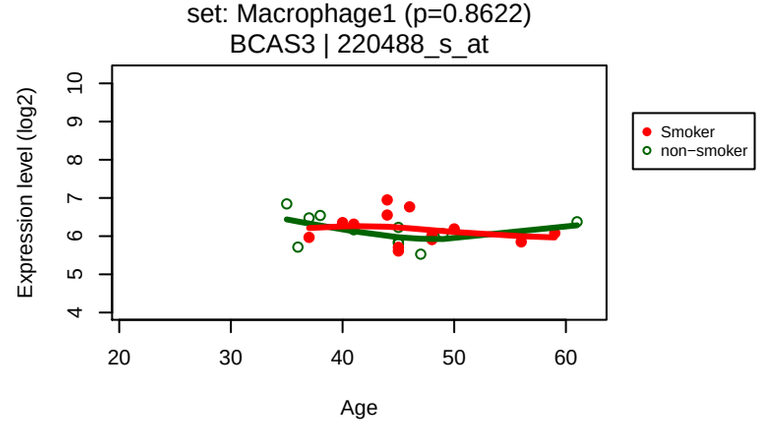

set: SmallAirway1 (p=0.2454)
BCAS3 | 220488_s_at
Expression level (log2)
Age
Smoker
non−smoker

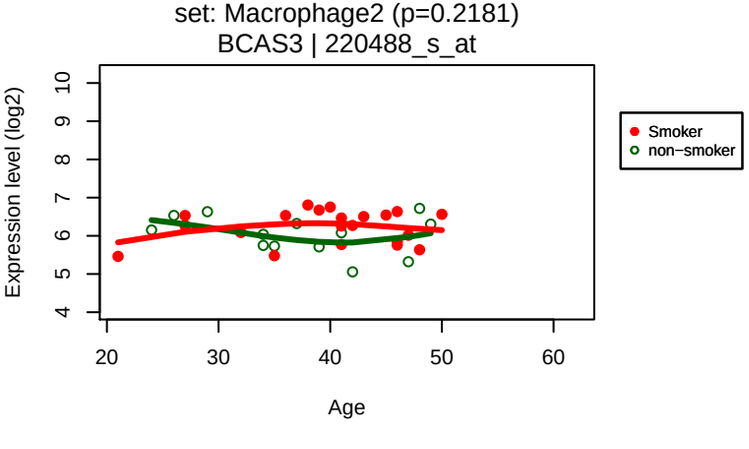

set: SmallAirway2 (p=0.1003)
BCAS3 | 220488_s_at
Expression level (log2)
Age
Smoker
non−smoker

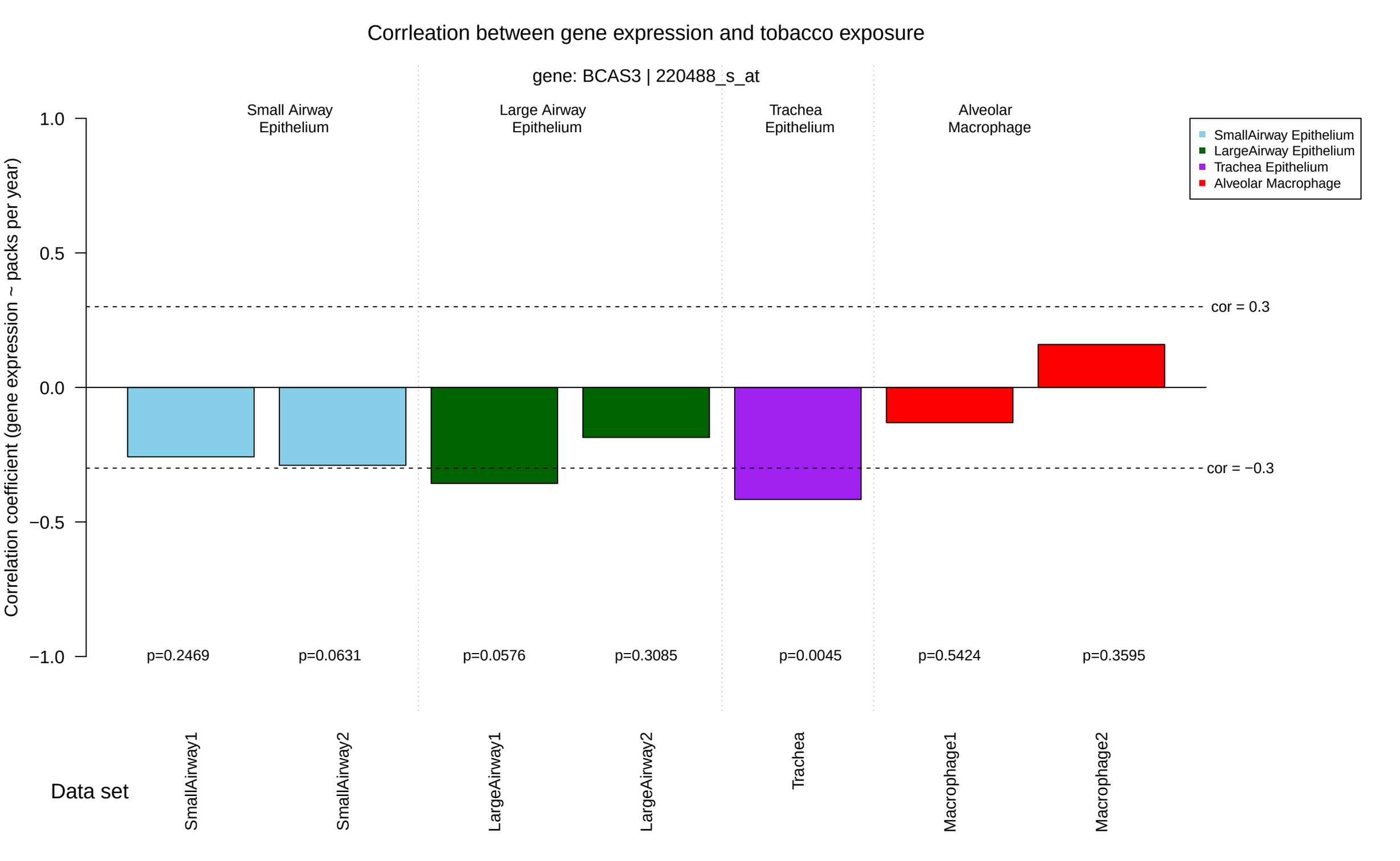

set: LargeAirway1 (p=0.0135)
BCAS3 | 220488_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9961)
BCAS3 | 220488_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0026)
BCAS3 | 220488_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8622)
BCAS3 | 220488_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2181)
BCAS3 | 220488_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: BCAS3 | 220488_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2469
p=0.0631
p=0.0576
p=0.3085
p=0.0045
p=0.5424
p=0.3595
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2